A Counterexample to Prescribed-Cycle Recovery in Barnette Graphs

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Abstract

Bekos, Kaufmann, and Pfister asked whether every prescribed Hamiltonian cycle of a Barnette graph can be recovered by choosing the dual spanning tree in the book-embedding algorithm of Alam et al. We answer this question negatively. An explicit 16-vertex Barnette graph has a Hamiltonian cycle whose complementary perfect matching meets all three edge-color classes in the simultaneous edge/face coloring used by the algorithm. Every output of the algorithm contains every edge designated green, independently of the spanning tree. The prescribed cycle omits a green edge under every global color labeling and therefore cannot be an output. We give a plane embedding, a two-expansion construction from the cube, a dependency-free verifier, and a pinned Lean companion that checks the principal finite proof interfaces.

1 Introduction

A *Barnette graph* is a simple, cubic, bipartite, 3-connected planar graph. Barnette's conjecture asserts that every such graph is Hamiltonian [2]. Although the conjecture remains open, several structural and computational results are known; in particular, every Barnette graph on at most 90 vertices is Hamiltonian [4]. Our graph is itself Hamiltonian and does not challenge that conjecture.

The question addressed here concerns a particular method for constructing long subhamiltonian cycles. Alam et al. used a simultaneous three-coloring of the edges and faces of an embedded Barnette graph to obtain a two-page book embedding [1]. Bekos, Kaufmann, and Pfister refined that construction and proved that it always produces a subhamiltonian cycle containing at least $5n/6$ graph edges [3]. Their Open Problem 2 asks whether, given a Barnette graph together with a particular Hamiltonian cycle, a choice of the spanning tree $\mathcal{T}^*_{bg}$ can make the algorithm produce that cycle [3, Open Problem 2, p. 6:6].

We give a negative answer. The obstruction is deliberately elementary: one necessary property of every algorithmic output is incompatible with one explicit Hamiltonian cycle. The certificate is robust under the possible interpretations of the color labels in the question. Even if the simultaneous coloring may be relabeled before the spanning tree is chosen, the obstruction persists.

Theorem 1. *There is an embedded Barnette graph G and a Hamiltonian cycle H of G such that, for every admissible simultaneous edge/face three-coloring and every choice of $\mathcal{T}^*_{bg}$, the construction of Alam et al. does not produce the cyclic vertex order of H.*

No prior resolution of this prescribed-cycle question was found in a documented search through August 11, 2026. This finite observation cannot exclude private or unindexed work.

2 The Algorithmic Obstruction

We recall only the part of the construction needed here. For an embedded Barnette graph G, the faces and edges admit simultaneous colors red, blue, and green with the following properties [1, 3]: adjacent faces have different colors; an edge incident with faces of colors x and y receives the third color z; the boundary of each face alternates between its two edge colors; and the dual subgraph induced by the blue and green faces is connected. The algorithm chooses a spanning tree $\mathcal{T}_{bg}^*$ of that dual subgraph and traverses it to construct a subhamiltonian cycle $\mathcal{C}$.

For a cyclic order π of $V(G)$, let $E_G(\pi)$ be the graph edges whose endpoints are consecutive in π, including the closing pair. Thus, if π is the cyclic order of a Hamiltonian cycle H, then $E_G(\pi) = E(H)$. The following necessary condition is Property 1 in the published presentation of Bekos et al.; it follows from their Invariant I.4 and is independent of the spanning tree [3, Property 1, p. 6:3]:

$$\text{every green edge of } G \text{ belongs to } E_G(\mathcal{C}). \tag{1}$$

For a Hamiltonian cycle H in a cubic graph, the edge set $M = E(G) \setminus E(H)$ is a perfect matching. Equation (1) immediately gives the following obstruction.

Lemma 2. *Suppose that, for every admissible simultaneous coloring of an embedded Barnette graph, the complementary matching M of a Hamiltonian cycle H contains a green edge. Then no choice of $\mathcal{T}_{bg}^*$ makes the algorithm produce the cyclic order of H.*

Proof. Choose a green edge $e \in M$. If the output order $\mathcal{C}$ were the cyclic order of H, then (1) would give $e \in E_G(\mathcal{C}) = E(H)$. This contradicts $e \in M = E(G) \setminus E(H)$, regardless of the chosen spanning tree. $\square$

3 The 16-Vertex Certificate

Let G have vertex set $\{0, 1, \ldots, 15\}$ and the adjacency lists in Table 1. We write $u\,v$ for the edge $\{u, v\}$, retaining the thin space between endpoint labels. Each edge occurs twice in the table, so it also gives an economical complete specification of G.

The following facial cycles specify a plane embedding:

$$\begin{aligned}
F_0 &= (0, 8, 9, 3, 5, 4), & F_1 &= (0, 15, 14, 1, 11, 8), \\
F_2 &= (0, 4, 12, 15), & F_3 &= (1, 14, 13, 7), \\
F_4 &= (1, 7, 6, 2, 10, 11), & F_5 &= (2, 6, 5, 3), \\
F_6 &= (2, 3, 9, 10), & F_7 &= (4, 5, 6, 7, 13, 12), \\
F_8 &= (8, 11, 10, 9), & F_9 &= (12, 13, 14, 15).
\end{aligned}$$

Every edge occurs in exactly two of these cycles and in opposite directions. The induced darts form one cyclic vertex link at each vertex, and the dual is connected. Together with $16 - 24 + 10 = 2$, these checks certify a connected cellular sphere embedding.

There is also a short construction certificate. Start from the cube on $0, \ldots, 7$, with facial cycles $(0, 1, 2, 3)$ and $(0, 1, 7, 4)$. First apply a C_4-expansion to the opposite edges $0\,3$ and $1\,2$: replace $0\,3$ by the path $0{-}8{-}9{-}3$, replace $1\,2$ by the path $1{-}11{-}10{-}2$, and add the cross edges $8\,11$ and $9\,10$. Then apply the same operation to $0\,1$ and $4\,7$: replace $0\,1$ by the path $0{-}15{-}14{-}1$, replace $4\,7$ by the path $4{-}12{-}13{-}7$, and add the cross edges $15\,12$ and $14\,13$. These are instances of the standard C_4-expansion used in the generation of Barnette graphs [5, 3], and they reconstruct the adjacency

Table 1: The graph G.

v	$N(v)$	v	$N(v)$
0	$4, 8, 15$	8	$0, 9, 11$
1	$7, 11, 14$	9	$3, 8, 10$
2	$3, 6, 10$	10	$2, 9, 11$
3	$2, 5, 9$	11	$1, 8, 10$
4	$0, 5, 12$	12	$4, 13, 15$
5	$3, 4, 6$	13	$7, 12, 14$
6	$2, 5, 7$	14	$1, 13, 15$
7	$1, 6, 13$	15	$0, 12, 14$

table exactly. The defining properties do not rest on that construction certificate: the table gives simplicity and cubicity, the displayed faces give planarity, and the bipartition is

$$\{0, 2, 5, 7, 9, 11, 12, 14\} \,\dot\cup\, \{1, 3, 4, 6, 8, 10, 13, 15\}\,.$$

For each of the 137 vertex sets D with $|D| \leq 2$, the graph $G - D$ is connected. Hence G is a Barnette graph. The verifier checks each assertion independently.

Consider the Hamiltonian cycle

$$H = (0, 4, 5, 3, 2, 6, 7, 13, 12, 15, 14, 1, 11, 10, 9, 8, 0). \tag{2}$$

Its complementary perfect matching is

$$M = \{0\,15, 1\,7, 2\,10, 3\,9, 4\,12, 5\,6, 8\,11, 13\,14\}\,. \tag{3}$$

Figure 1 shows the plane graph, with the matching edges distinguished by color class.

It remains to verify the simultaneous coloring. One proper face coloring is

$$\begin{array}{r|l} 0 & F_0, F_4, F_9 \\ 1 & F_1, F_6, F_7 \\ 2 & F_2, F_3, F_5, F_8. \end{array} \tag{4}$$

The induced edge-color classes are

$$\begin{aligned} E_0 &= \{0\,15, 1\,14, 2\,3, 4\,12, 5\,6, 7\,13, 8\,11, 9\,10\}\,, \\ E_1 &= \{0\,4, 1\,7, 2\,6, 3\,5, 8\,9, 10\,11, 12\,15, 13\,14\}\,, \\ E_2 &= \{0\,8, 1\,11, 2\,10, 3\,9, 4\,5, 6\,7, 12\,13, 14\,15\}\,. \end{aligned}$$

The facial boundary conditions and the required dual connectedness follow directly from the face list. In particular, an edge in E_i is incident with faces carrying the other two colors. These conditions are checked edge by edge in both proof companions.

Lemma 3. *The embedding has exactly six proper face three-colorings. They are the six global permutations of* (4)*.*

Proof. The three faces F_0, F_1, F_2 are pairwise adjacent around vertex 0, so their colors determine a permutation of $0, 1, 2$. Fix their colors as in (4). Then F_8 is forced by F_0, F_1; F_6 by F_0, F_8; F_5 by F_0, F_6; F_4 by F_1, F_5; F_3 by F_1, F_4; F_7 by F_4, F_5; and F_9 by F_2, F_7. No choice remains. Permuting the initial three colors gives all six colorings. □

Finally, (3) gives

$$
\begin{aligned}
M \cap E_0 &= \{0\,15, 4\,12, 5\,6, 8\,11\}\,,\\
M \cap E_1 &= \{1\,7, 13\,14\}\,,\\
M \cap E_2 &= \{2\,10, 3\,9\}\,.
\end{aligned}
$$

All three intersections are nonempty. By Lemma 3, every admissible relabeling designates one of these classes green. Therefore H omits a green edge under every labeling, and Lemma 2 proves Theorem 1.

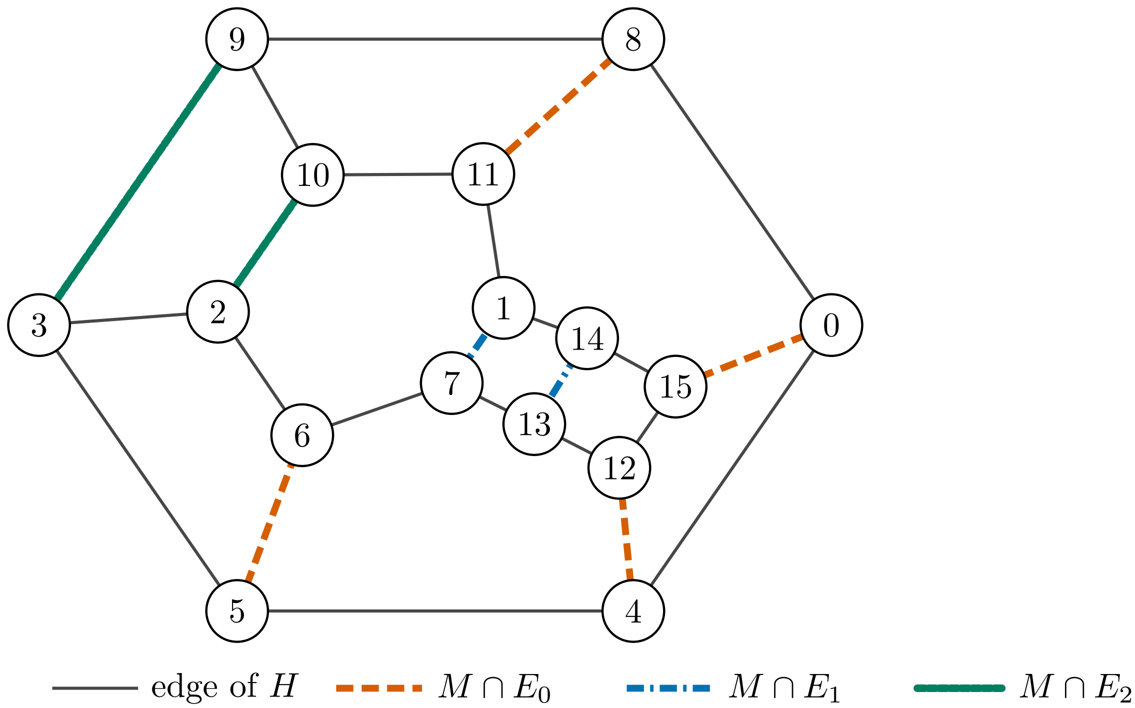

Figure 1: The prescribed Hamiltonian cycle H (solid) and its complementary matching M. Line color and dash pattern distinguish the three induced edge-color classes; every class contains an edge of M.

4 Verification and Scope

The supplementary verifier uses only the Python standard library. It checks the graph and all displayed data directly: degrees, bipartiteness, 3-connectivity, oriented facial incidences, Euler characteristic, vertex links, the two C_4-expansions, Hamiltonicity of (2), the complementary matching, all six face colorings, the induced edge classes, the five coloring conditions used by the source, and the three nonempty intersections above. It does not encode or simulate the spanning-tree traversal, because Equation (1) is already a tree-independent necessary condition.

The pinned Lean companion checks the finite obstruction and its proof interfaces. It proves that every displayed cycle step is an edge, including the closing step, and that the cycle edges and omitted matching are an exact disjoint partition of $E(G)$. It encodes all oriented facial boundary darts and their incidences, checks two opposite face incidences per edge, derives the displayed dual from shared primal edges, and proves that the face-color boundary unions induce exactly the three displayed edge-color classes. It also checks all two-color dual connectedness conditions, uniqueness of the face coloring up to global permutation, all three matching intersections, and connectivity after every deletion of at most two vertices (137 cases). Finally, it proves the abstract forced-green contradiction.

The finite propositions use `native_decide`, so this artifact explicitly trusts Lean's code generator in addition to the kernel; the source includes `#print axioms` audits. Lean does not formalize the topological theorem interpreting the oriented incidence data as a cellular sphere embedding, nor does it reprove Property 1 or the full book-embedding algorithm. Those boundaries are recorded with the formal source.

The result concerns universality of one graph-drawing/book-embedding construction. It neither supplies a non-Hamiltonian Barnette graph nor weakens the evidence for Barnette's conjecture. Instead, it shows that a Hamiltonian Barnette graph may have a Hamiltonian cycle lying outside the output family of that construction.

Data and Code Availability

The complete JSON certificate, dependency-free verifier, pinned Lean source and manifest, and captured verification outputs are archived with the preprint at `https://doi.org/10.5281/zenodo.21890733`.

Declaration of Interests

The author declares no competing interests and no external funding for this work.

Declaration of Agentic AI and AI-Assisted Technologies

During the discovery, verification, and preparation of this work, the author used OpenAI Codex and Anthropic Claude as agentic research assistants. Codex assisted with literature and novelty searches, certificate verification, code testing, proof development, and manuscript drafting and editing. Claude was used for independent adversarial review, including attempts to identify mathematical, computational, citation, and presentation defects. The author independently checked the claims, proof, computation, sources, and final text, accepts full responsibility for the work, and confirms that neither system is an author.

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